

UNIT 2 – GAUSS’S LAW

2.2 Electric Flux and Gauss’s Law: Spherical Symmetry**Answer key**

Placeholder worksheet for the site prototype. Free for classroom use.

1. A closed surface encloses point charges $q_1 = +2.0$ nC and $q_2 = -5.0$ nC (and a third charge of $+7.0$ nC sits outside). Find the net electric flux through the surface.

Only enclosed charge counts: $\Phi_E = \frac{q_{\text{enc}}}{\epsilon_0} = \frac{-3.0 \times 10^{-9}}{8.85 \times 10^{-12}} = -3.4 \times 10^2 \text{ N} \cdot \text{m}^2/\text{C}.$

2. An insulating sphere of radius R carries charge Q spread uniformly through its volume. Use Gauss’s law to find $E(r)$ for $r < R$ and for $r > R$.

Gaussian sphere of radius r : $E(4\pi r^2) = q_{\text{enc}}/\epsilon_0$. Outside, $q_{\text{enc}} = Q$ so $E = \frac{kQ}{r^2}$. Inside, $q_{\text{enc}} = Q \frac{r^3}{R^3}$ so $E = \frac{kQr}{R^3}$. The two agree at $r = R$.